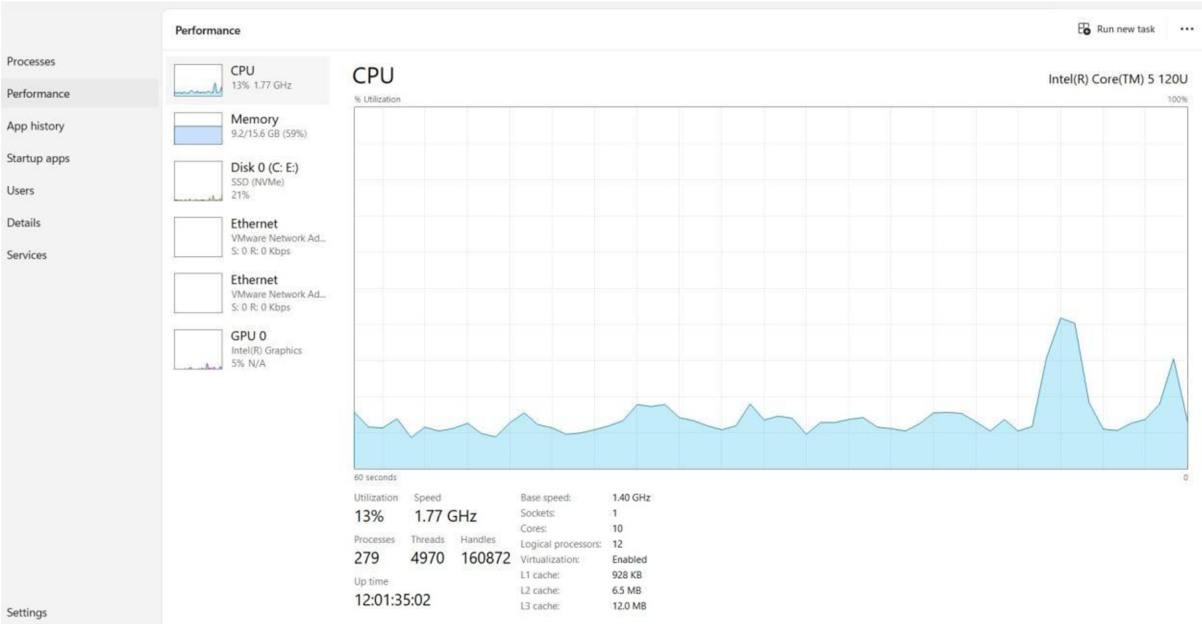
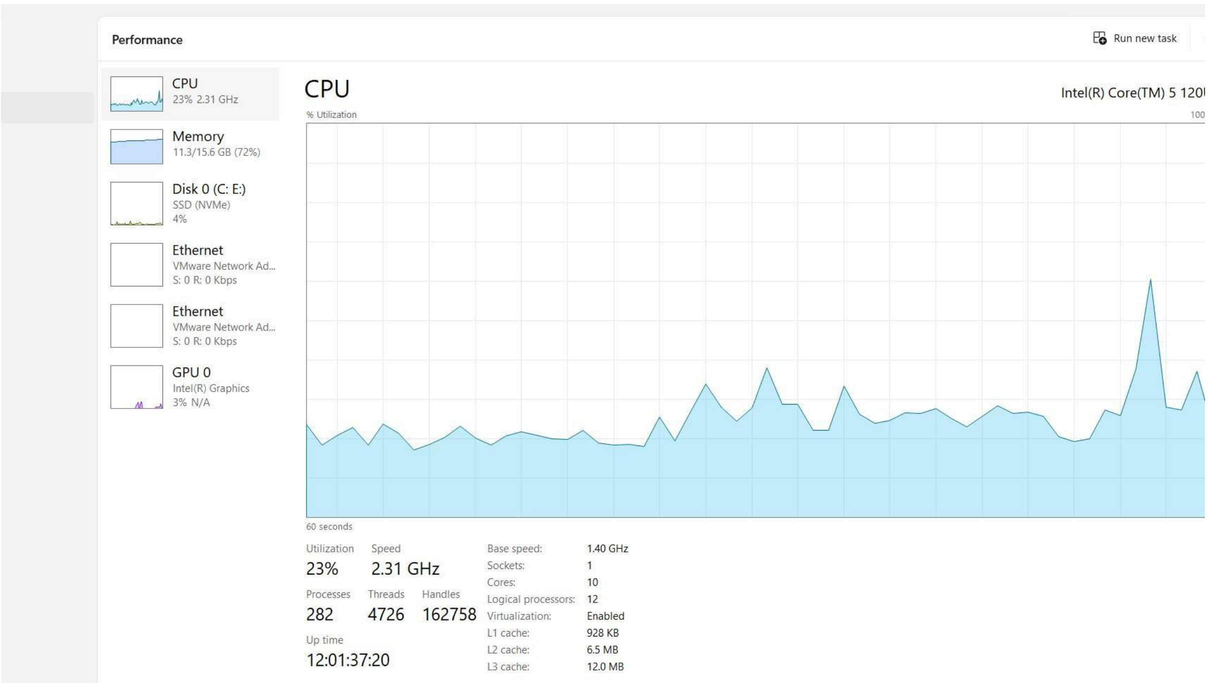
Experiment 24(VMware /virtual box):

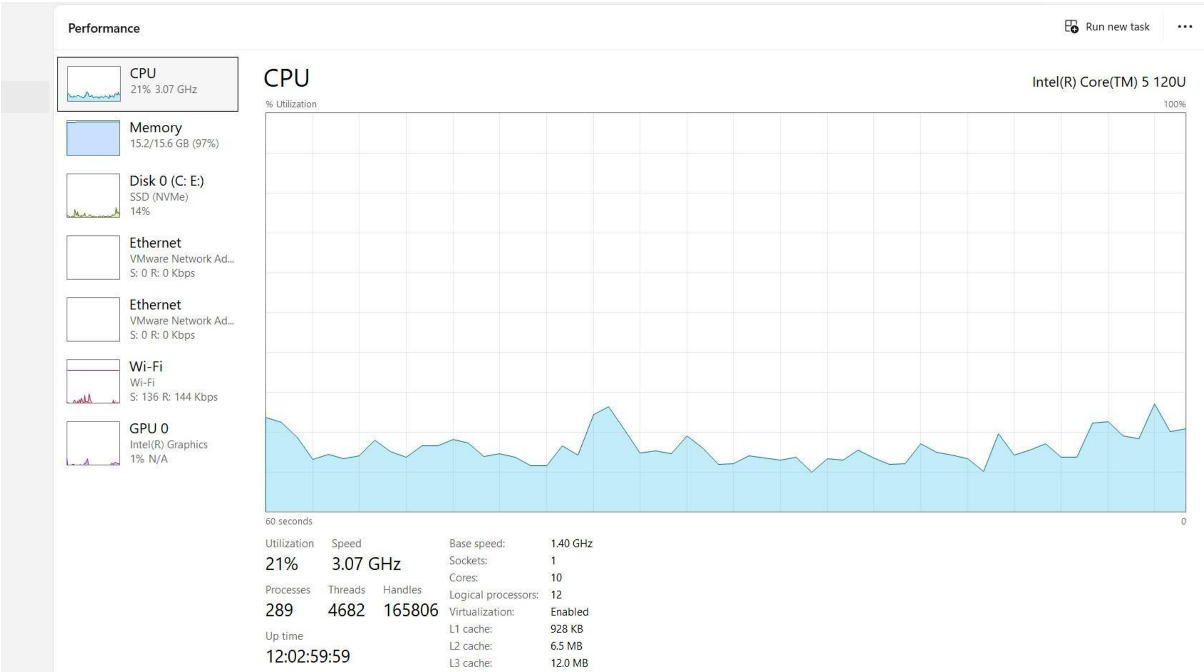
Before open VM :



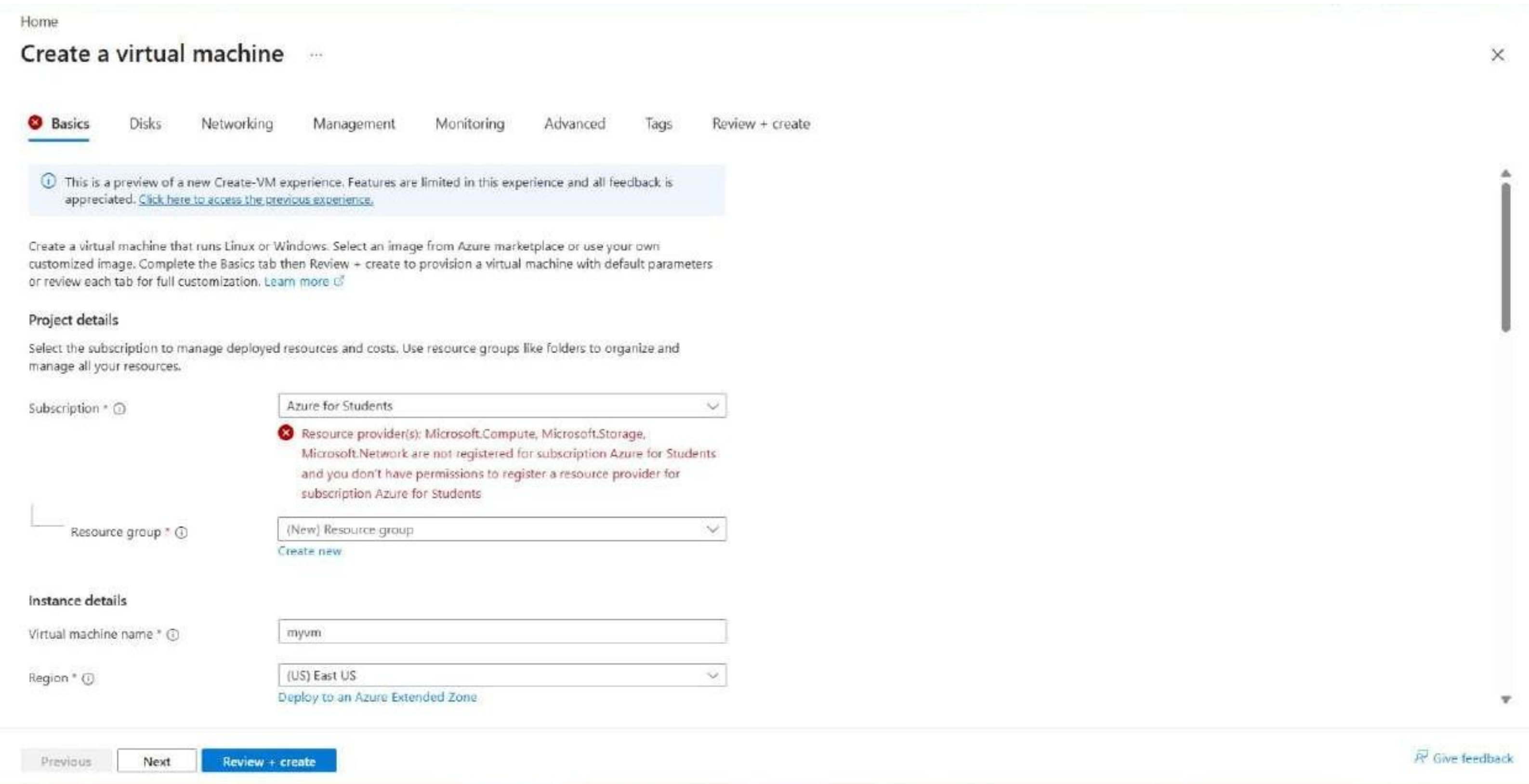
After open one VM shows the value of memory =72%

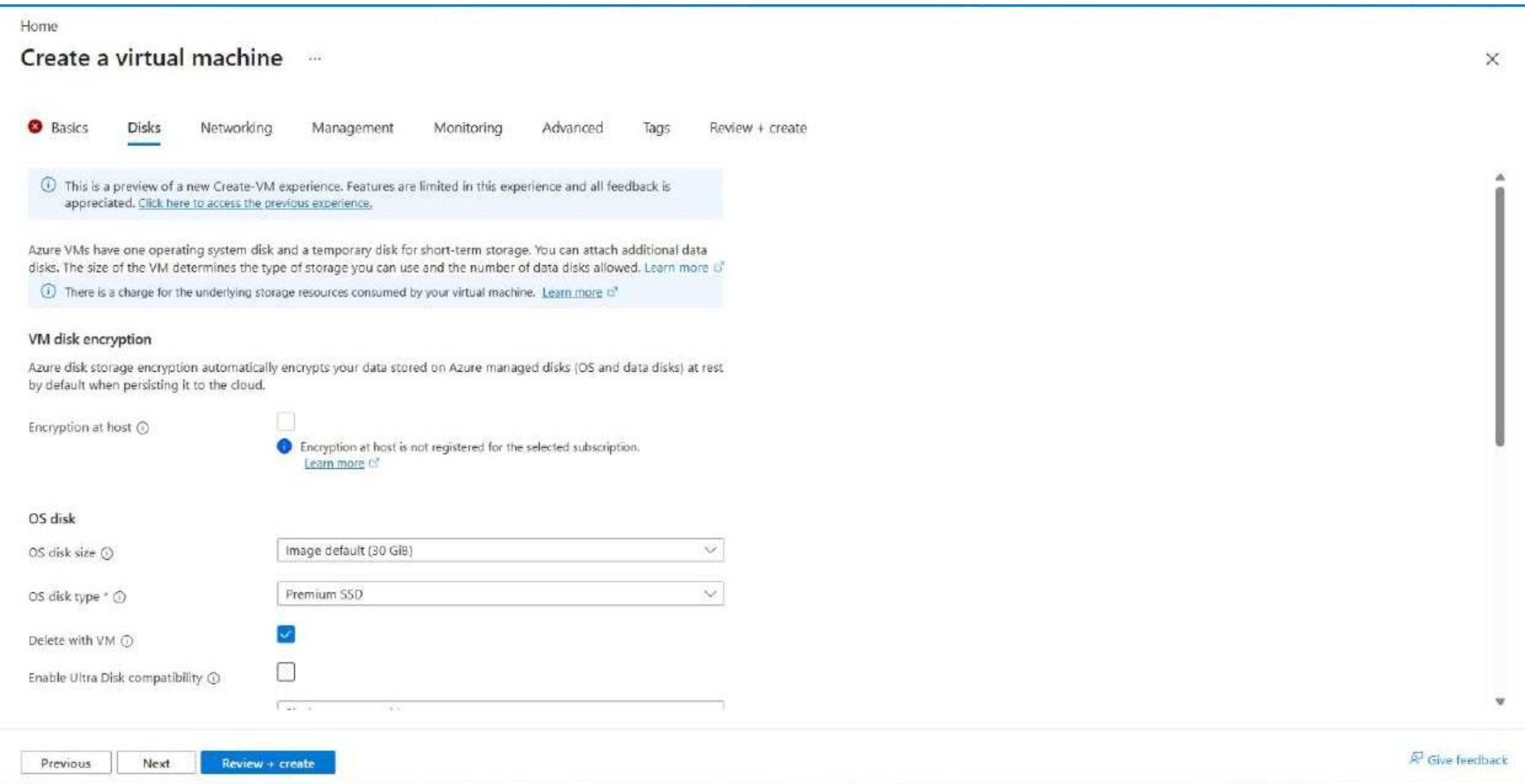


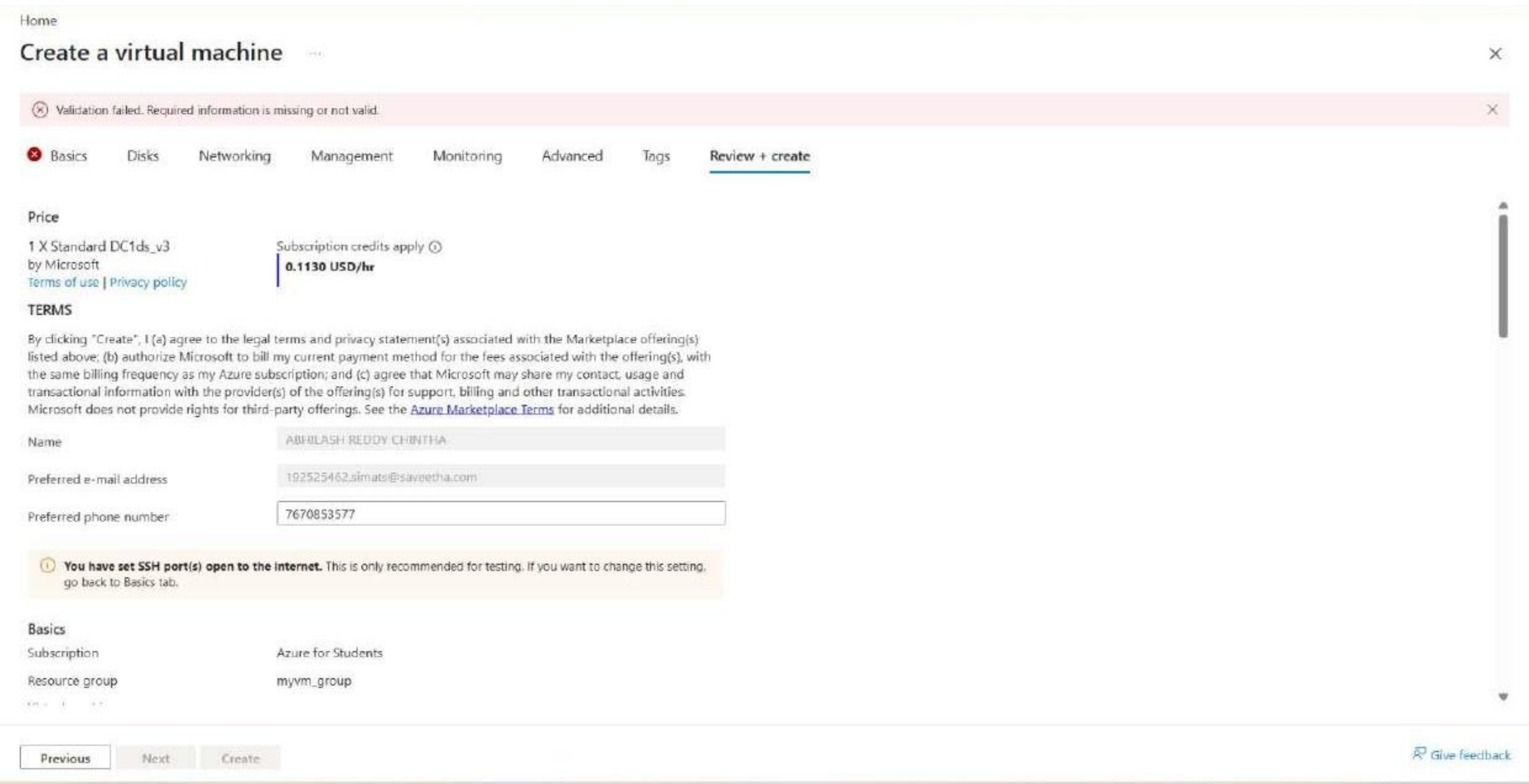
After opening 2 VM shows the value of memory=97%



EXP -25,26,27:


Home
Create a virtual machine
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
This is a preview of a new Create-VM experience. Features are limited in this experience and all feedback is appreciated. Click here to access the previous experience.
Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. Learn more
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure for Students
Resource provider(s): Microsoft.Compute, Microsoft.Storage, Microsoft.Network are not registered for subscription Azure for Students and you don't have permissions to register a resource provider for subscription Azure for Students
Resource group *
(New) Resource group
Create new
Instance details
Virtual machine name *
myvm
Region *
(US) East US
Deploy to an Azure Extended Zone
Previous
Next
Review + create
Give feedback


Home
Create a virtual machine
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
This is a preview of a new Create-VM experience. Features are limited in this experience and all feedback is appreciated. Click here to access the previous experience.
Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. Learn more
There is a charge for the underlying storage resources consumed by your virtual machine. Learn more
VM disk encryption
Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.
Encryption at host
Encryption at host is not registered for the selected subscription.
Learn more
OS disk
OS disk size
Image default (30 GiB)
OS disk type *
Premium SSD
Delete with VM
Enable Ultra Disk compatibility
Previous
Next
Review + create
Give feedback


Home
Create a virtual machine
Validation failed. Required information is missing or not valid.
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
Price
1 X Standard DC1ds_v3
by Microsoft
Terms of use | Privacy policy
Subscription credits apply
0.1130 USD/hr
TERMS
By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the Azure Marketplace Terms for additional details.
Name
ABHILASH REDDY CHINTHA
Preferred e-mail address
192525462.simats@saveetha.com
Preferred phone number
7670853577
You have set SSH port(s) open to the internet. This is only recommended for testing. If you want to change this setting, go back to Basics tab.
Basics
Subscription
Azure for Students
Resource group
myvm_group
Previous
Next
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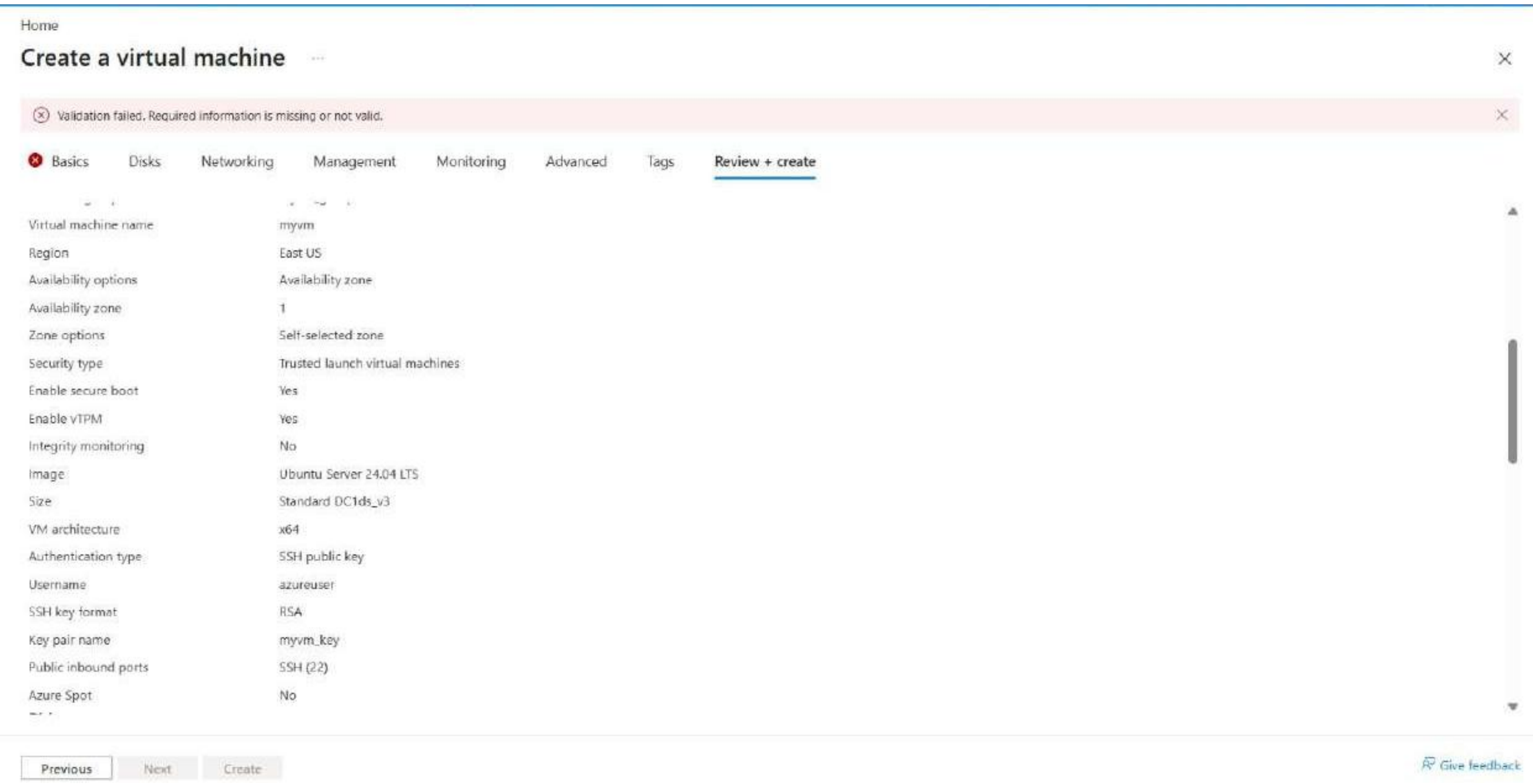
Give feedback


Home
Create a virtual machine
Validation failed. Required information is missing or not valid.
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
Virtual machine name
myvm
Region
East US
Availability options
Availability zone
Availability zone
1
Zone options
Self-selected zone
Security type
Trusted launch virtual machines
Enable secure boot
Yes
Enable vTPM
Yes
Integrity monitoring
No
Image
Ubuntu Server 24.04 LTS
Size
Standard DC1ds_v3
VM architecture
x64
Authentication type
SSH public key
Username
azureuser
SSH key format
RSA
Key pair name
myvm_key
Public inbound ports
SSH (22)
Azure Spot
No
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Home

Create a virtual machine

Validation failed. Required information is missing or not valid.

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | **Review + create**

Disks

OS disk size	Image default
OS disk type	Premium_LRS
Use managed disks	Yes
Delete OS disk with VM	Yes
Ephemeral OS disk	None

Networking

Virtual network	(new) vnet-eastus-1
Subnet	(new) snet-eastus-1
Public IP	(new) myvm-ip
NIC Network security group	-
Accelerated networking	Off
Place this virtual machine behind an existing load balancing solution?	No
Delete public IP and NIC when VM is deleted	Disabled

Management

System assigned managed identity	Off
Login with Microsoft Entra ID	Off
Backup	Off

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Login with Microsoft Entra ID	Off
Backup	Off
Site Recovery	Off
Enable periodic assessment	Off
Enable hotpatch	Off
Patch orchestration options	Image default
Auto-shutdown	Off
Enable hibernation	Off
Monitoring	
Alerts	Off
Boot diagnostics	On
Enable OS guest diagnostics	Off
Enable application health monitoring	Off
Advanced	
Extensions	None
VM applications	None
Cloud init	No
Capacity reservation group	None
Disk controller type	SCSI

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Give feedback

EXP_28-37:

Exp-1 Execute a Python Program in Google Cloud Shell

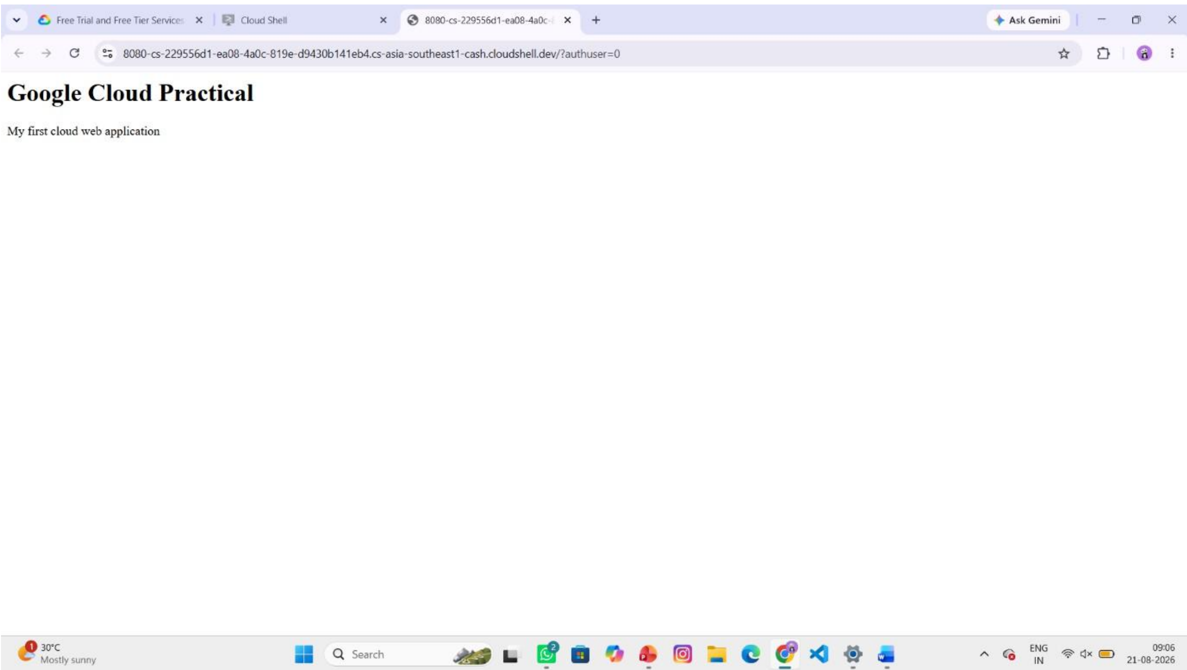
```
Welcome to Cloud Shell! Type "help" to get started.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
akulareddy072007@cloudshell:~$ nano hello.py
akulareddy072007@cloudshell:~$ python3 hello.py
Hello from Google Cloud!
Welcome to Cloud Computing
```

Exp-2 Create and Manage Files in Cloud Storage of Cloud Shell

```
Welcome to Cloud Computing
akulareddy072007@cloudshell:~$ mkdir cloudlab
mkdir: cannot create directory 'cloudlab': File exists
akulareddy072007@cloudshell:~$ cd cloudlab
akulareddy072007@cloudshell:~/cloudlab$ echo "Welcome to Google Cloud" > sample.txt
akulareddy072007@cloudshell:~/cloudlab$ cat sample.txt
Welcome to Google Cloud
akulareddy072007@cloudshell:~/cloudlab$ ls
hello.py  index.html  sample.txt
akulareddy072007@cloudshell:~/cloudlab$ cp sample.txt backup.txt
akulareddy072007@cloudshell:~/cloudlab$ ls
backup.txt  hello.py  index.html  sample.txt
akulareddy072007@cloudshell:~/cloudlab$
```

Exp-3. Run a Simple Web Server in Google Cloud Shell

Exp-4. System Resource Monitoring in Google Cloud Shell



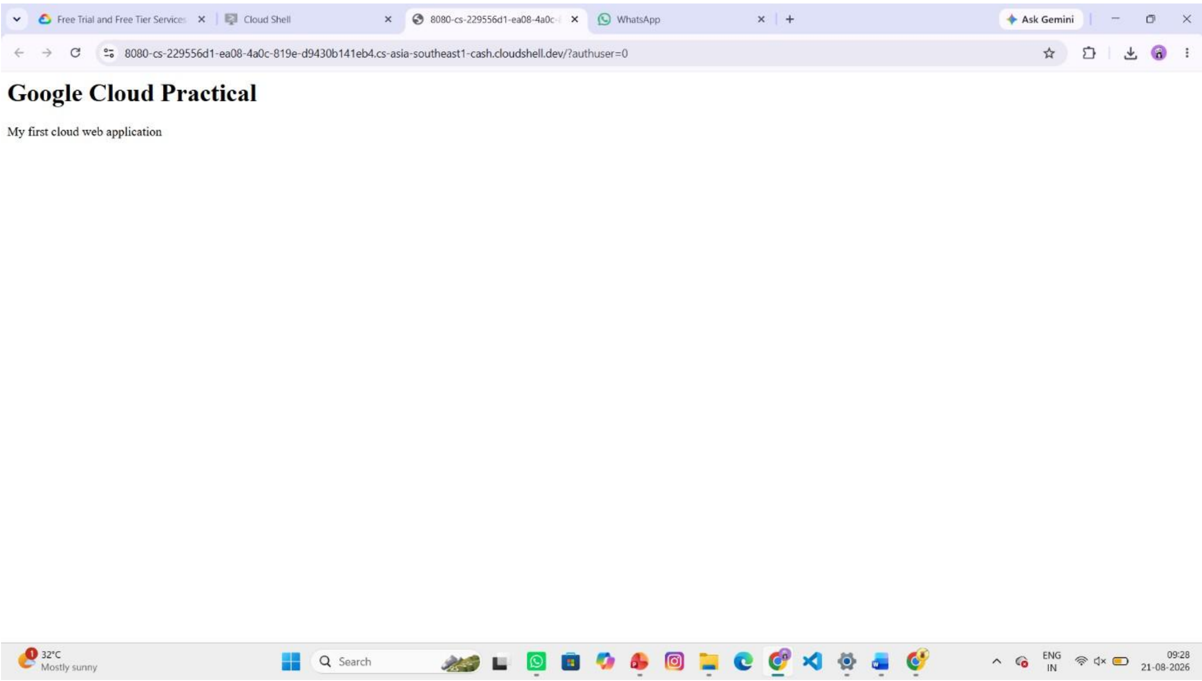
```
top - 03:53:25 up  2:07,  0 user,  load average: 0.00, 0.05, 0.28
Tasks:  45 total,   1 running,  44 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.0 us,  0.8 sy,  0.0 ni, 98.0 id,  0.0 wa,  0.0 hi,  0.2 si,  0.1 st
MiB Mem :  15998.9 total,  11892.6 free,   2446.6 used,   1938.5 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.  13552.3 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1638 akulare+  20   0 1475372  83728  48592 S   0.3   0.5   0:04.59 MainThread
   2975 akulare+  20   0   18.5g   1.0g  63892 S   0.3   6.4   0:41.54 MainThread
   4070 akulare+  10 -10   11852   5864   3684 R   0.3   0.0   0:00.01 top
      1 root      20   0    4324   3288   3004 S   0.0   0.0   0:00.07 bash
      9 syslog    20   0  222140   3836   2480 S   0.0   0.0   0:00.06 rsyslogd
     25 root      20   0   38796  28148  10912 S   0.0   0.2   0:01.05 python
     26 root      20   0    6196   2712   2472 S   0.0   0.0   0:00.00 logger
     75 root      10 -10   12020   4048   2936 S   0.0   0.0   0:00.05 sshd
     79 root      10 -10   12016   2864   1760 S   0.0   0.0   0:00.00 sshd
    374 root      20   0 2099112  88084  64116 S   0.0   0.5   0:08.71 dockerd
    451 root      20   0 1269404  13804   8516 S   0.0   0.1   0:01.15 editor-proxy
    452 root      20   0   16328   6332   5408 S   0.0   0.0   0:00.01 sudo
    465 root      20   0 1228132   3064   2200 S   0.0   0.0   0:00.00 tmux-agent
    505 root      20   0 1894952  42512  29756 S   0.0   0.3   0:04.85 containerd
    696 root      20   0    2696   1580   1484 S   0.0   0.0   0:00.00 sleep
    750 akulare+  10 -10    7544   3088   2028 S   0.0   0.0   0:00.24 tmux
    751 akulare+  10 -10    9312   6296   3928 S   0.0   0.0   0:00.04 bash
   1146 root      20   0    9196   4160   3780 S   0.0   0.0   0:00.00 runuser
   1147 akulare+  20   0    4324   3376   3108 S   0.0   0.0   0:00.00 sh
   1169 akulare+  20   0 1781032 148336  52568 S   0.0   0.9   0:11.66 MainThread
   1650 akulare+  20   0    8920   6100   4116 S   0.0   0.0   0:00.03 bash
   2024 akulare+  10 -10    9312   6476   4108 S   0.0   0.0   0:00.04 bash
   2476 akulare+  10 -10    9312   6424   4028 S   0.0   0.0   0:00.11 bash
   2951 akulare+  10 -10  104004  20696  10780 S   0.0   0.1   0:00.41 python3
   2960 root      10 -10   13992   9784   8404 S   0.0   0.1   0:00.01 sshd
   2962 root      10 -10   13992   9716   8344 S   0.0   0.1   0:00.01 sshd
   2964 akulare+  10 -10   14252   6368   4712 S   0.0   0.0   0:00.00 sshd
   2965 akulare+  10 -10   14252   6444   4788 S   0.0   0.0   0:00.00 sshd
   2966 akulare+  10 -10    7340   3640   3388 S   0.0   0.0   0:00.00 bash
   2969 akulare+  10 -10    7604   4492   3896 S   0.0   0.0   0:00.00 bash
   2972 akulare+  10 -10    7340   3596   3348 S   0.0   0.0   0:00.00 bash
   2973 akulare+  10 -10    7340   3688   3428 S   0.0   0.0   0:00.00 start-shell.sh
   2974 akulare+  10 -10    6876   3472   3172 S   0.0   0.0   0:00.00 tmux
   2987 akulare+  20   0 1448728  63964  41904 S   0.0   0.4   0:00.33 MainThread
```

Exp-5. Execute a C Program in Google Cloud Shell

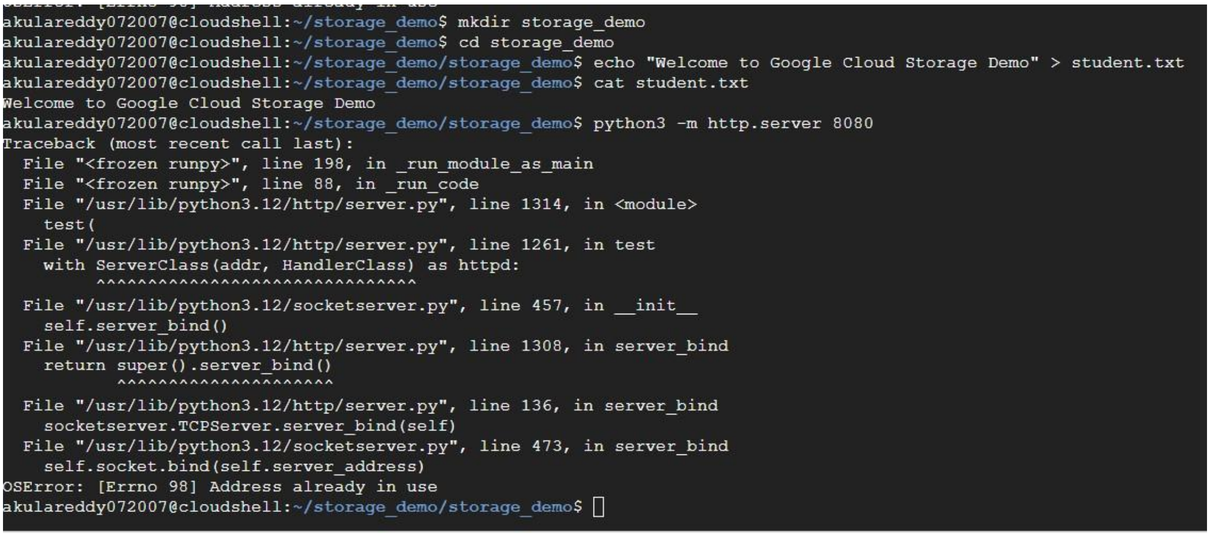
```
akulareddy072007@cloudshell:~$ nano add.c
akulareddy072007@cloudshell:~$ ./add
-bash: ./add: No such file or directory
akulareddy072007@cloudshell:~$ gcc add.c -o add
akulareddy072007@cloudshell:~$ nano add.c
akulareddy072007@cloudshell:~$ ./add
Sum = 30
akulareddy072007@cloudshell:~$
```

Exp-**6. Simple Python Web Application – PaaS**

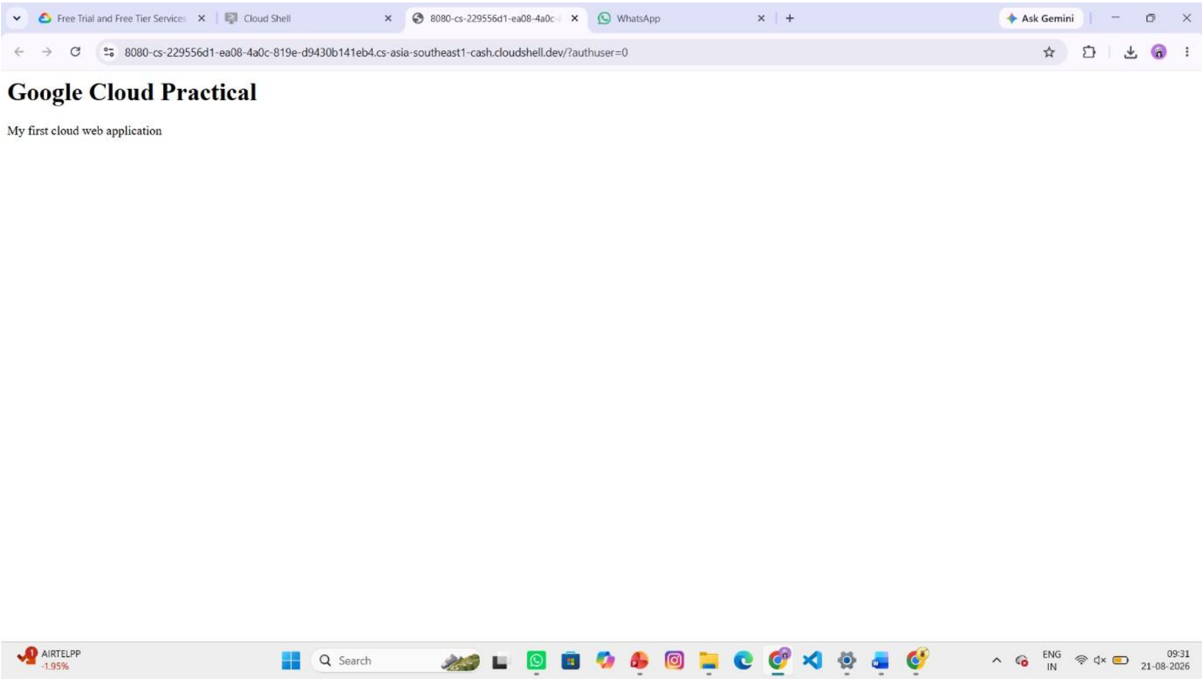
```
Welcome to Cloud Shell! Type "help" to get started.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
akulareddy072007@cloudshell:~$ nano app.py
akulareddy072007@cloudshell:~$ python3 app.py
Traceback (most recent call last):
  File "/home/akulareddy072007/app.py", line 7, in <module>
    HTTPServer(("", 8080), MyHandler).serve_forever()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/socketserver.py", line 457, in __init__
    self.server_bind()
  File "/usr/lib/python3.12/http/server.py", line 136, in server_bind
    socketserver.TCPServer.server_bind(self)
  File "/usr/lib/python3.12/socketserver.py", line 473, in server_bind
    self.socket.bind(self.server_address)
OSError: [Errno 98] Address already in use
akulareddy072007@cloudshell:~$ 
```



exp-**7. Storage as a Service – Store and Access a File**



```
akulareddy072007@cloudshell:~/storage_demo$ mkdir storage_demo
akulareddy072007@cloudshell:~/storage_demo$ cd storage_demo
akulareddy072007@cloudshell:~/storage_demo/storage_demo$ echo "Welcome to Google Cloud Storage Demo" > student.txt
akulareddy072007@cloudshell:~/storage_demo/storage_demo$ cat student.txt
Welcome to Google Cloud Storage Demo
akulareddy072007@cloudshell:~/storage_demo/storage_demo$ python3 -m http.server 8080
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "/usr/lib/python3.12/http/server.py", line 1314, in <module>
    test(
  File "/usr/lib/python3.12/http/server.py", line 1261, in test
    with ServerClass(addr, HandlerClass) as httpd:
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/socketserver.py", line 457, in __init__
    self.server_bind()
  File "/usr/lib/python3.12/http/server.py", line 1308, in server_bind
    return super().server_bind()
           ^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/http/server.py", line 136, in server_bind
    socketserver.TCPServer.server_bind(self)
  File "/usr/lib/python3.12/socketserver.py", line 473, in server_bind
    self.socket.bind(self.server_address)
OSError: [Errno 98] Address already in use
akulareddy072007@cloudshell:~/storage_demo/storage_demo$
```



Exp-**8. SQL Storage + Basic Query – DaaS**

```
OSError: [Errno 98] Address already in use
akulareddy072007@cloudshell:~/storage_demo/storage_demo$ sqlite3 student.db
SQLite version 3.45.1 2024-01-30 16:01:20
Enter ".help" for usage hints.
sqlite> CREATE TABLE student(
 id INTEGER,
 name TEXT
);
sqlite>
sqlite> INSERT INTO student VALUES(1,'Anu');
INSERT INTO student VALUES(2,'Priya');
sqlite> SELECT * FROM student;
1|Anu
2|Priya
sqlite>
```